

Explainability of Intracranial Haemorrhage Detection Across Clinical CT Backbones: A Comparative Evaluation of ResNet-50, ConvNeXt-Base, and ViT-B/16

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Abstract— Forensic pathology services face heavy caseloads and limited specialist capacity. Post-mortem CT (PMCT) also remains scarce and costly to annotate compared with large clinical CT archives. This suggests a natural training strategy: build intracranial haemorrhage (ICH) detectors on abundant clinical data instead. However, classification accuracy alone does not establish whether a model's reasoning can be trusted in a high-stakes setting. This study asks whether a clinically trained ICH detection model can provide explanations that are faithful and stable enough to be useful when a forensic professional reviews a scan, without requiring expensive pixel-level annotations. Three backbones trained on RSNA ICH under a shared split and input representation are compared: ResNet-50, ViT-B/16, and ConvNeXt-Base. Beyond classification, the study evaluates the faithfulness (AOPC) and stability (Max-Sensitivity) of their Grad-CAM explanations, performs a bone-window correlation check, and calibrates all three models. The results do not point to one architecture that is best on every dimension. ResNet-50's epidural detection and explanation faithfulness are both weak ($F1 = 0.176$, $AOPC = 0.114 \pm 0.059$). ViT-B/16 has a much stronger epidural AOPC (0.505 ± 0.206) and the best macro classification metrics. ConvNeXt-Base gives the strongest AOPC and Max-Sensitivity results for intraparenchymal and intraventricular haemorrhage. The explanation results also do not line up cleanly with the visual examples or the bone-channel check. Clinical training provides a useful starting point, but explanation faithfulness and stability have to be measured rather than assumed. The bone-channel check adds a separate test for spatial association with the bone window. The results therefore support a comparison of tradeoffs between the three backbones rather than a single architecture ranking. The next stage is evaluation under PMCT domain shift.

Keywords— *Intracranial Haemorrhage Detection, Explainable AI, Grad-CAM, Vision Transformer, ConvNeXt, ResNet, Model Calibration, Forensic Radiology, Post-Mortem CT*

I. INTRODUCTION

Forensic pathology departments, particularly in high-caseload settings such as India, routinely face more post-mortem imaging than available specialists can review in depth. Post-mortem CT (PMCT) is increasingly used to support medico-legal death investigation [1]. PMCT datasets suitable for training or validating machine learning models remain small, however, and pixel-level annotation of haemorrhage in PMCT is slow and expensive.

Clinical CT offers a way around this bottleneck. The RSNA ICH Detection Challenge dataset [2] provides hundreds of thousands of labelled slices, enough to train intracranial haemorrhage (ICH) detectors at scale. Matsumoto et al. [3] showed that models trained on this kind of clinical data can be transferred to PMCT for classification; they compared fifteen transfer-learned models and found clinical-to-PMCT transfer feasible. Their work leaves open a separate question. It does not establish whether the explanations attached to these classifiers, which is what a forensic reviewer would actually

inspect, remain faithful and useful once the model leaves the clinical domain.

A classifier's output alone gives a forensic professional little to act on. If a model's attribution points to the wrong region of a scan, the explanation could mislead rather than help a reviewer. That makes explanation faithfulness worth testing alongside classification. The first problem is deciding what “faithful” means operationally. AOPC provides one answer by testing whether the regions Grad-CAM ranks highly actually affect the model's output. That is useful, but it does not establish whether the highlighted region is the haemorrhage. A model can depend strongly on a region for the wrong reason, so a good perturbation score is evidence about model dependence rather than evidence of anatomical correctness.

Grad-CAM [4] is a practical way to produce visual attributions, but the map is still a post-hoc description of model behaviour, so it needs an independent check. Plain Grad-CAM rather than HiResCAM [5] is used throughout

this study because it applies consistently across convolutional and transformer backbones, which keeps the cross-architecture comparison even. Two separate questions are then asked about the resulting maps. AOPC tests whether the regions ranked highly by the explanation actually matter to the model's prediction. Max-Sensitivity tests whether the map changes when the input is perturbed. Neither metric, by itself, establishes that the highlighted anatomy is the correct cause of the prediction.

Three architecturally distinct backbones are compared, spanning a conventional CNN, a transformer, and a modernised CNN that borrows transformer-era design choices: ResNet-50 [6], ViT-B/16 [7], and ConvNeXt-Base [8]. The aim is not to assume that one architecture should be best. It is to determine whether architecture changes the relationship between classification performance and explanation behaviour, and whether one configuration provides a clearly stronger basis for the later PMCT domain-shift evaluation.

The work reported here is a clinical-CT baseline. The PMCT stage, evaluating these models and their explanations under domain shift, has not been performed and is described only as future work in Section IX. Nothing here should be read as evidence of clinical-to-PMCT transfer, forensic usefulness, or forensic validation. The main question at this stage is narrower: what does each backbone actually do on the source-domain data, and where do the explanation metrics agree or disagree with the classification results?

II. RESEARCH QUESTION AND CONTRIBUTIONS

A. Research Question

Can a clinically trained ICH detection model provide sufficiently faithful, useful, and architecture-robust explanations to support forensic professionals reviewing post-mortem CT, without requiring expensive pixel-level annotations?

B. Comparative Sub-Question

Given three backbone and training configurations evaluated on RSNA, which provides the strongest overall evidence across classification performance, explanation faithfulness (AOPC), explanation stability (Max-Sensitivity), calibration, and the bone-channel correlation check? No aggregation method is defined in the source methodology, so no single combined score is computed and no formal overall rank is assigned. Instead, the architectures are compared metric by metric, and the pattern of tradeoffs is used to determine which models should be carried into the PMCT evaluation.

C. Contributions

1. A matched-split, matched-input benchmark of ResNet-50, ViT-B/16, and ConvNeXt-Base on RSNA ICH.

2. Quantitative validation of Grad-CAM explanations using AOPC and Max-Sensitivity across all five haemorrhage subtypes and the aggregate “any” label.
3. A bone-window shortcut sanity check.
4. Calibration analysis at the aggregate and subtype level.
5. A metric-by-metric comparison of the evidence that can inform selection of candidate backbones for the pending PMCT evaluation.

III. RELATED WORK

Automated ICH detection from CT has been studied extensively since the RSNA challenge dataset [2] established a large, multiply-labelled benchmark for haemorrhage subtype classification. ResNet-50 [6] and ViT-B/16 [7] represent two architecturally distinct approaches to the underlying vision problem: one convolutional and one transformer-based. ConvNeXt [8] revisits convolutional design with training and scaling choices closer to modern transformers. Its inclusion here is deliberate. It lets the comparison span a conventional CNN, a pure transformer, and a modernised CNN built on transformer-era design principles, rather than treating ConvNeXt-Base as a second, redundant convolutional baseline.

Class Activation Mapping and its descendants remain common tools for producing post-hoc visual attributions from these models. CAM [9] required architectural constraints that Grad-CAM [4] later removed, which is why plain Grad-CAM, rather than a CAM variant that assumes a particular architecture, is used here for cross-backbone consistency. Perturbation-based faithfulness metrics such as AOPC [10] and robustness metrics such as Max-Sensitivity [11] give a way to quantify explanation quality beyond visual inspection, and both are used in this study. Precision-Recall analysis is reported alongside ROC-AUC [12] because it is more informative than ROC-AUC alone under class imbalance, which matters given how rare epidural haemorrhage is in this dataset. Calibration is assessed following standard post-hoc temperature scaling [13].

Two clinical references anchor the anatomical and PMCT context used in Sections IV and IX: an overview of intracranial haemorrhage imaging [14], and studies documenting post-mortem CT artefacts that can be mistaken for pathology [15, 16]. These sources inform the domain-shift hypotheses discussed later rather than the classification methodology itself.

IV. MATERIALS AND METHODS

A. Dataset and Split

The dataset used is the RSNA Intracranial Haemorrhage Detection challenge set [2].

TABLE I Dataset Split

Split	n
Train	188,402
Validation	40,372
Test	40,373

The split is 70/15/15, stratified on the binary “any” label, with random_state = 42. The validation set was used for early stopping, per-class F1-optimal thresholds, and temperature-scaling calibration. The held-out test set was used for all reported metrics and was not touched during tuning.

TABLE II Test-Set Positive Counts (Multi-Label, n = 40,373)

Label	Positives
any	16,104
epidural	477 ($\approx 1.18\%$)
intraparenchymal	5,494
intraventricular	3,980
subarachnoid	5,361
subdural	6,985

B. Input Representation

Each CT slice was rendered as a three-channel image using three clinical CT windows. Each channel carries independently meaningful windowed information, and is not a cosmetic colour map applied over greyscale CT.

TABLE III Windowing Applied per Input Channel

TABLE IV Training Configuration by Backbone

	ResNet-50	ViT-B/16	ConvNeXt-Base
Pretrained weights	ResNet50_Weights.IMAGENET1K_V2	ViT_B_16_Weights.IMAGENET1K_V1	ImageNet1K weights (per notebook)
Unfrozen depth	layer4	final encoder layer only (encoder.layers[-1])	last stage only (features[-1])
Optimizer	SGD, momentum 0.9	AdamW	AdamW
Learning rate	1e-3	1e-4	1e-4
Weight decay	n/a	1e-4	1e-4
Warmup	none	1 epoch, linear, factor 0.1 $\rightarrow$ base LR	1 epoch, linear, factor 0.1 $\rightarrow$ base LR
Scheduler	Cosine annealing	Cosine annealing after warmup	Cosine annealing after warmup
Batch / max epochs / patience	32 / 15 / 5	32 / 15 / 5	32 / 15 / 5
Loss	BCEWithLogitsLoss	BCEWithLogitsLoss	BCEWithLogitsLoss
Best checkpoint	epoch 14, val loss 0.1905	epoch 5, val loss 0.1726 (vit_b16_best.pt)	epoch 13, val loss 0.1759

An earlier reporting summary stated that ViT-B/16 unfroze its last two encoder blocks; the configuration above reflects the corrected, single-block setting used for training.

Optimizer, learning rate, and warmup differ across the three configurations, and fine-tuned depth also differs. This is best read as a comparison of three complete, reasonable training setups. It is not a controlled ablation that isolates architecture as the only variable.

D. Explainability Methodology

Channel	Window	W / L	Purpose
R	Brain	W80/L40	Soft-tissue contrast
G	Subdural	W200/L80	Wider blood/brain contrast
B	Bone	W2000/L600	Skull/bone detail

ImageNet normalisation was applied throughout. Training used RandomHorizontalFlip(); evaluation did not. A BalancedRandomSampler equalised the marginal sampling probability of any=1 and any=0 examples in each training epoch, so $P_S(\text{any}=1) = P_S(\text{any}=0) = 0.5$. It did not reweight the subtype distribution conditional on any=1, so $P_S(\text{subtype} | \text{any}=1) = P_data(\text{subtype} | \text{any}=1)$. Epidural therefore remained the least represented subtype within the positive class. With an epidural prevalence of $\approx 1.18\%$ in the held-out test set, even a modest number of false positives produces a substantial precision penalty. The sampler reduces background dominance from the negative class, but it does not balance the five haemorrhage subtypes.

Because the bone window is part of the input, a Grad-CAM map could in principle track skull intensity rather than haemorrhage. This possibility motivated the bone-channel correlation check reported in Section V-D.

C. Backbones and Training Configurations

All three models start from ImageNet pretraining with most layers frozen, then are fine-tuned with a new Dropout(0.3) $\rightarrow$ Linear head.

Grad-CAM: Grad-CAM was applied to all three backbones via pytorch_grad_cam, using plain Grad-CAM rather than HiResCAM for cross-architecture consistency [4, 5]. The target layers were layer4[-1] for ResNet-50, features[-1][-1] for ConvNeXt-Base, and encoder.layers[-1].ln_1 for ViT-B/16. For ViT-B/16, the reshape transform removes the CLS token and reshapes the remaining 196 patch tokens into a 14 $\times$ 14 spatial grid before Grad-CAM is computed. CAMs were computed on confirmed true positives (ground-truth positive and model-positive at the validation-derived

threshold), sampled without replacement with `random_state = 42`.

$$AOPC_c(x) = 1/K \sum_{k=1}^K [f_c(x)^{(0)} - f_c(x)^{(k)}], K = 10$$

Here, $f_c(x)^{(0)}$ is the unperturbed class probability and $f_c(x)^{(k)}$ is the probability after the k -th cumulative perturbation. The protocol uses a fixed 16×16 patch grid over the 224×224 input, with each selected patch replaced by its per-channel image mean. The protocol is identical across backbones: 50 confirmed true positives per class where available (epidural had 127 available for ResNet-50, of which 50 were sampled), with 10 cumulative perturbation steps. Each selected patch was replaced with its per-channel image mean, chosen because zero is a meaningful intensity value within a windowed CT channel rather than a neutral baseline. ConvNeXt-Base's true final feature map is 7×7 at this input size, but the same 16×16 perturbation grid was used for protocol consistency across backbones.

The Grad-CAM figures provide concrete examples of how the three backbones differ, but they are not a population-level sample. As shown in the representative ViT-B/16 visualisations (Fig. 1), several maps are diffuse or extend towards peripheral skull margins. ResNet-50 is more mixed, and its two displayed epidural false negatives show different failure modes (Fig. 3). ConvNeXt-Base more often concentrates around plausible haemorrhage regions (Fig. 5). The confirmed ResNet-50 epidural true positives can show attention near the skull-brain interface, but that does not offset the much weaker quantitative epidural AOPC and classification results.

ViT-B/16 achieves the highest epidural AOPC (0.5047 ± 0.2058), yet several of its displayed attributions extend into peripheral or skull-associated regions (Fig. 1). This discrepancy highlights a divergence between perturbation sensitivity and qualitative localisation.

Max-Sensitivity: This metric measures the sensitivity of the Grad-CAM map to small Gaussian perturbations of the input. Let $CAM(x)$ denote the normalised Grad-CAM map, and let

δ_i be the i -th noise realisation. The reported Max-Sensitivity is given by (2).

$$Max-Sensitivity(x) = \max_{i=1,\dots,N} \|CAM(x + \delta_i) - CAM(x)\|_2, \delta_i \sim N(0, r^2 I), r = 0.05\sigma_x, N = 10$$

Here, σ_x is the standard deviation of the image values, and $r = 0.05\sigma_x$ sets the Gaussian noise scale for each Monte Carlo sample. The perturbed inputs were clipped to $[0, 1]$, Grad-CAM was recomputed for each of the $N = 10$ realisations, and the maximum L2 difference from the original CAM was reported. Lower values indicate greater stability under the specified perturbation.

Bone-channel correlation: This check tests whether the spatial pattern of each Grad-CAM map is associated with the bone-window channel. A strong correlation would be a reason to inspect whether skull intensity is influencing the explanation, which matters because bone information is explicitly included in the input. Pearson correlation only measures spatial association between two flattened maps; it cannot establish that the model used bone as a causal shortcut, and a low correlation cannot rule out more indirect dependence. Correlations were recorded as NaN where either vector had negligible variance.

E. Calibration

Temperature scaling was fit on the validation set and evaluated on the held-out test set, using ECE, MCE, Brier score, and NLL, reported both in aggregate and per subtype. Subtype-level calibration is reported separately because a single global temperature can improve overall probability estimates while making a rare subtype worse. That distinction becomes important for epidural haemorrhage in the results below.

V. RESULTS

A. Classification

Table V reports per-class ROC-AUC, PR-AUC, decision threshold, precision, recall, specificity, F1, and positive counts for each backbone.

TABLE V Per-Class Classification Performance

Model	Class	ROC-AUC	PR-AUC	Thresh.	Prec.	Recall	Spec.	F1	Pos.
ResNet-50	any	0.9497	0.9345	0.5471	0.8528	0.8539	0.9022	0.8533	16,104
	epidural	0.8703	0.0944	0.0883	0.1319	0.2662	0.9790	0.1764	477
	intraparenchymal	0.9469	0.8231	0.3490	0.7937	0.7110	0.9709	0.7501	5,494
	intraventricular	0.9665	0.8250	0.3693	0.7503	0.7389	0.9731	0.7446	3,980
	subarachnoid	0.8923	0.6440	0.3402	0.5831	0.5814	0.9363	0.5822	5,361
	subdural	0.9172	0.7558	0.3794	0.6772	0.6863	0.9316	0.6817	6,985
ViT-B/16	any	0.9572	0.9451	0.4586	0.8691	0.8655	0.9135	0.8673	16,104
	epidural	0.9361	0.3970	0.2772	0.3966	0.4340	0.9921	0.4144	477
	intraparenchymal	0.9556	0.8481	0.4941	0.8294	0.7015	0.9773	0.7601	5,494

	intraventricular	0.9687	0.8310	0.2956	0.7539	0.7337	0.9738	0.7437	3,980
	subarachnoid	0.9165	0.7155	0.3516	0.6489	0.6348	0.9474	0.6418	5,361
	subdural	0.9346	0.7985	0.3862	0.7028	0.7382	0.9347	0.7201	6,985
ConvNeXt-Base	any	0.9566	0.9405	0.4883	0.8351	0.8914	0.8832	0.8623	16,104
	epidural	0.9291	0.2513	0.1704	0.2652	0.3836	0.9873	0.3136	477
	intraparenchymal	0.9556	0.8436	0.4357	0.7988	0.7117	0.9718	0.7527	5,494
	intraventricular	0.9748	0.8546	0.4372	0.7759	0.7645	0.9739	0.7702	3,980
	subarachnoid	0.9066	0.6699	0.3435	0.5978	0.6251	0.9356	0.6111	5,361
	subdural	0.9353	0.7945	0.4610	0.7092	0.7150	0.9387	0.7121	6,985

Macro ROC-AUC / macro PR-AUC were 0.9186 / 0.6284 for ResNet-50, 0.9423 / 0.7180 for ViT-B/16, and 0.9403 / 0.6828 for ConvNeXt-Base.

Epidural haemorrhage is the clearest example of why the classification metrics need to be read together. At 1.18% test prevalence, its ROC-AUC is much higher than its PR-AUC for all three models, which is exactly the kind of divergence expected when positives are rare. The architectures still separate meaningfully on the practical measures. ResNet-50 has epidural ROC-AUC 0.8703, PR-AUC 0.0944, and F1 0.1764. ViT-B/16 reaches 0.9361, 0.3970, and 0.4144, while

ConvNeXt-Base reaches 0.9291, 0.2513, and 0.3136. ViT-B/16 is therefore clearly ahead of ConvNeXt-Base on epidural PR-AUC and F1, not only on ROC-AUC. At the same time, the 477 positive cases are a much smaller evidence base than the positive counts for the other subtypes, so this one class is insufficient to define an overall architecture ranking. The shared any-based sampler also does not balance the five subtypes, which is a plausible contributor to rare-subtype behaviour. It cannot, by itself, explain why ResNet-50 falls so much further behind the other two.

B. Representative Grad-CAM Visualisations

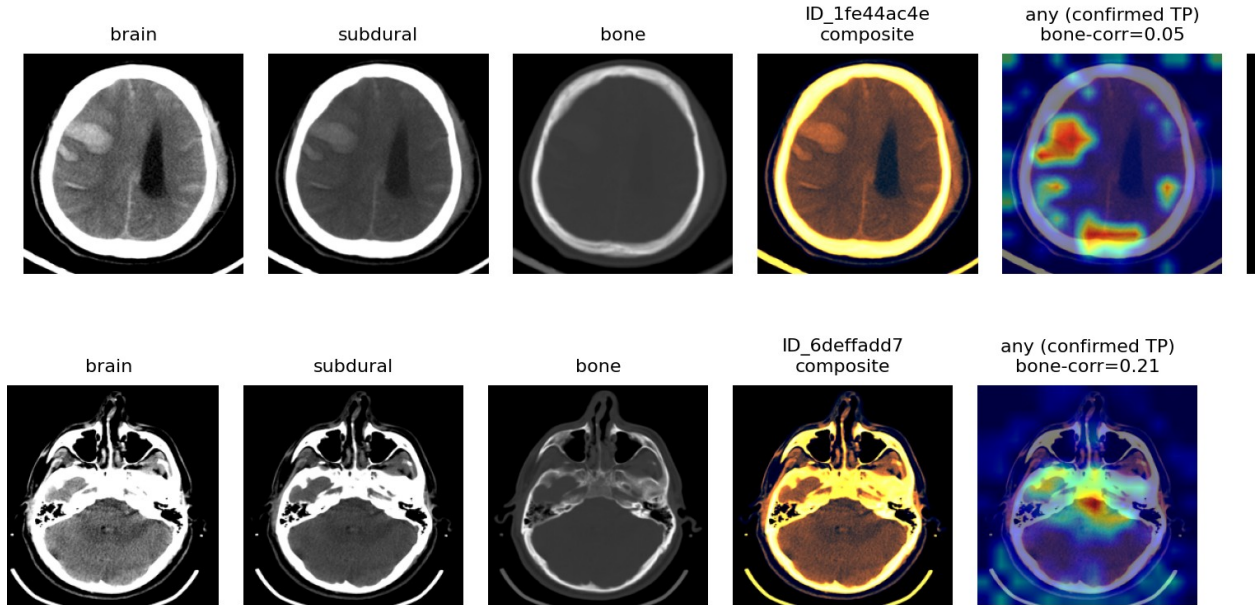


Fig. 1. Representative ViT-B/16 Grad-CAM visualisations across confirmed true-positive cases spanning haemorrhage classes. Several attributions are diffuse or extend toward peripheral skull margins.

Fig. 2. ViT-B/16 ROC and precision-recall curves for epidural haemorrhage on the test split.

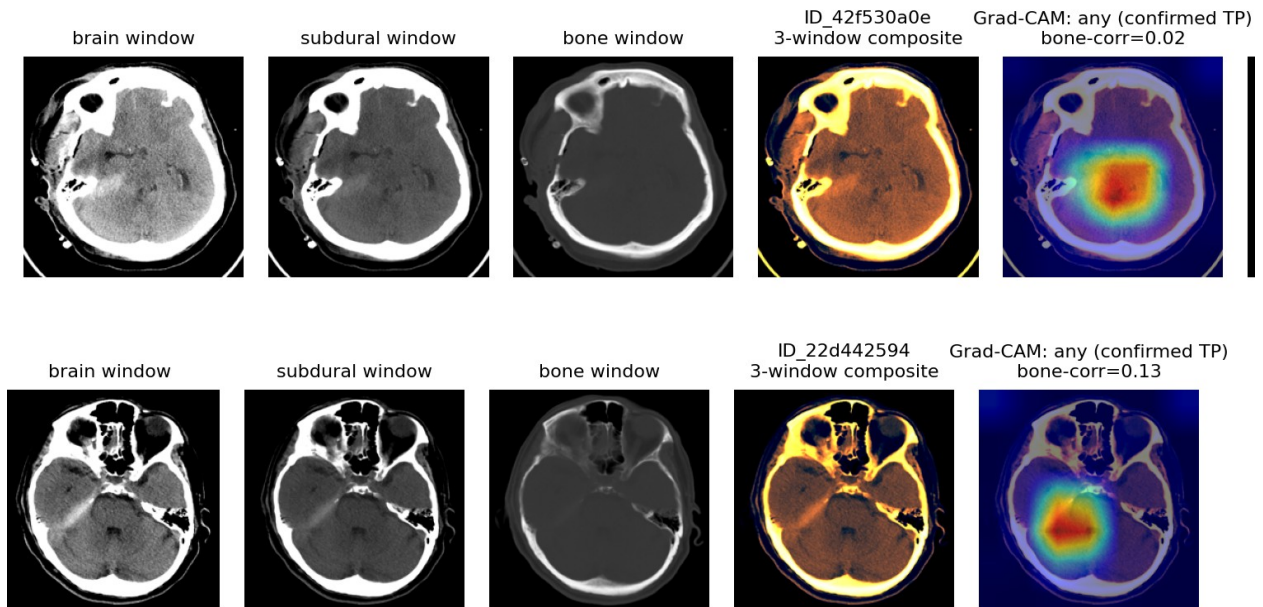
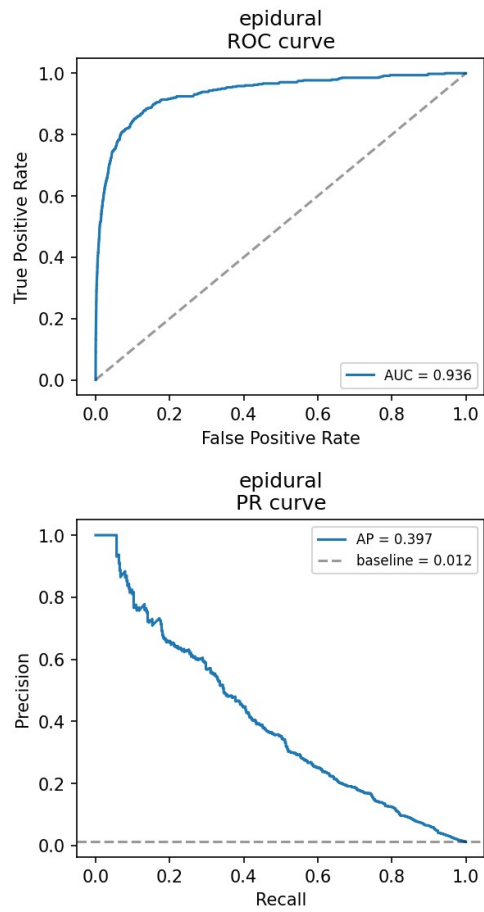


Fig. 3. Representative ResNet-50 Grad-CAM visualisations, including two epidural false negatives that show different failure modes.

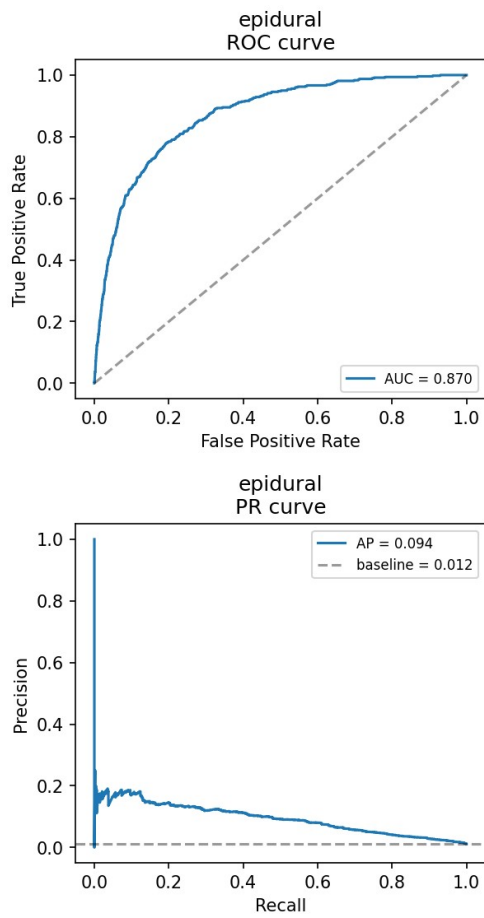


Fig. 4. ResNet-50 ROC and precision-recall curves for epidural haemorrhage, illustrating the weak precision-recall performance at 1.18% prevalence.

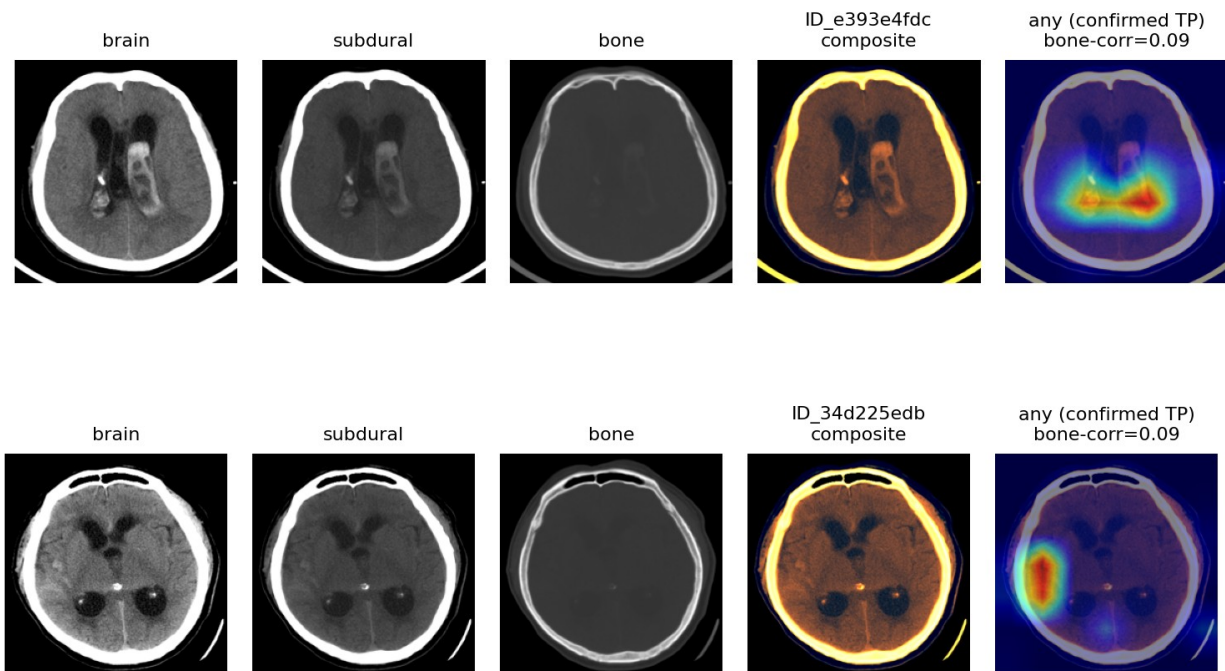


Fig. 5. Representative ConvNeXt-Base Grad-CAM visualisations, more often concentrated around plausible haemorrhage regions.

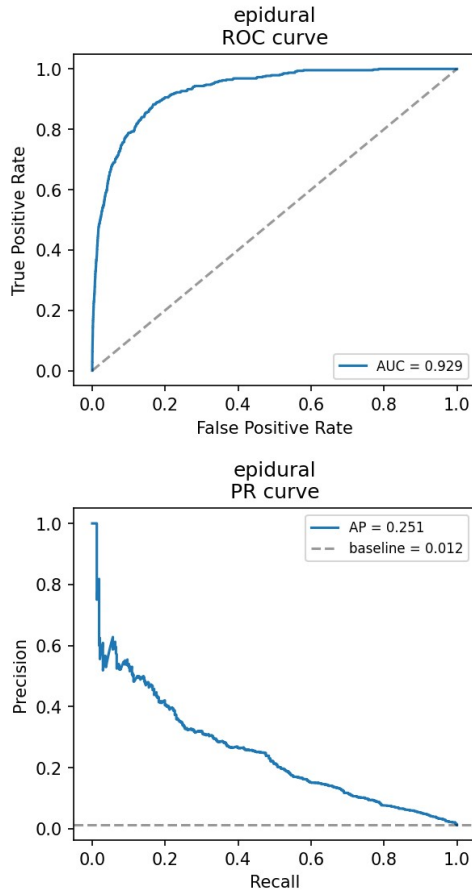


Fig. 6. ConvNeXt-Base ROC ($AUC = 0.929$) and precision-recall ($AP = 0.251$) curves for epidural haemorrhage against a prevalence baseline of 0.012.

Figures 1, 3, and 5 show representative Grad-CAM behaviour, while Figures 2, 4, and 6 show the corresponding epidural ROC and precision-recall curves. Figure 1 contains the diffuse and peripheral ViT-B/16 examples, whereas Figure 4 shows the particularly poor ResNet-50 precision-recall performance at the 1.18% epidural prevalence. Figures 2 and 6 provide the corresponding ViT-B/16 and ConvNeXt-Base classification curves. The figures are useful for identifying concrete failure modes. The cross-backbone comparison, however, is based primarily on the quantitative results, because the displayed cases are illustrative rather than a statistical sample.

C. AOPC and Max-Sensitivity

Table VI reports AOPC and Max-Sensitivity for $n = 50$ confirmed true positives per class.

TABLE VI AOPC and Max-Sensitivity by Backbone ($n = 50$ per Class)

Model	Class	AOPC (mean $\pm$ SD)	Max-Sens. (mean $\pm$ SD)
ResNet-50	any	0.2727 $\pm$ 0.1764	35.0383 $\pm$ 10.4159
	epidural	0.1140 $\pm$ 0.0588	32.6009 $\pm$ 9.3948
	intraparenchymal	0.7008 $\pm$ 0.1938	14.0190 $\pm$ 8.8351
	intraventricular	0.7304 $\pm$ 0.1927	21.4233 $\pm$ 12.3235
	subarachnoid	0.4609 $\pm$ 0.2053	30.2297 $\pm$ 14.0988
	subdural	0.5395 $\pm$ 0.1602	30.3747 $\pm$ 14.9248
ViT-B/16	any	0.3859 $\pm$ 0.1963	24.1945 $\pm$ 7.9772
	epidural	0.5047 $\pm$ 0.2058	29.9434 $\pm$ 7.5488
	intraparenchymal	0.6335 $\pm$ 0.2666	23.0017 $\pm$ 9.8101
	intraventricular	0.6942 $\pm$ 0.2253	21.9595 $\pm$ 7.5424
	subarachnoid	0.6274 $\pm$ 0.2102	21.6558 $\pm$ 8.3443
	subdural	0.5627 $\pm$ 0.2085	21.2431 $\pm$ 7.2466
ConvNeXt-Base	any	0.4597 $\pm$ 0.1872	20.1052 $\pm$ 13.9192
	epidural	0.3380 $\pm$ 0.1381	18.2150 $\pm$ 6.9497
	intraparenchymal	0.7725 $\pm$ 0.1988	7.3721 $\pm$ 6.2315
	intraventricular	0.7522 $\pm$ 0.1515	13.4200 $\pm$ 9.3295
	subarachnoid	0.5592 $\pm$ 0.2029	20.7250 $\pm$ 9.5121
	subdural	0.5255 $\pm$ 0.1661	17.6109 $\pm$ 8.8528

The AOPC results show several different patterns rather than one architecture dominating. The epidural result is the clearest separation: ViT-B/16 has 0.5047 ± 0.2058 , compared with 0.3380 ± 0.1381 for ConvNeXt-Base and 0.1140 ± 0.0588 for ResNet-50. The ViT-ResNet gap is larger than either model's SD, so the difference is substantial relative to the case-to-case spread in these samples. The ViT-ConvNeXt gap is much less decisive, because it is smaller than ViT's SD and close to ConvNeXt's SD. ViT-B/16's epidural AOPC advantage is therefore clear over ResNet-50 but less conclusive as a separation from ConvNeXt-Base.

For intraparenchymal haemorrhage, ConvNeXt-Base has the highest AOPC at 0.7725 ± 0.1988 , but the difference from ResNet-50's 0.7008 ± 0.1938 is smaller than either model's SD, and ViT-B/16 has an even larger spread at 0.6335 ± 0.2666 . The same issue appears for intraventricular AOPC, where ConvNeXt-Base is 0.7522 ± 0.1515 and ResNet-50 is 0.7304 ± 0.1927 . Those means favour ConvNeXt-Base, but the case-level variation makes the margin less decisive than the decimal values suggest.

Max-Sensitivity gives a stronger separation for intraparenchymal haemorrhage: ConvNeXt-Base is 7.3721 ± 6.2315 , versus 14.0190 ± 8.8351 for ResNet-50 and 23.0017 ± 9.8101 for ViT-B/16. Even here, the SDs show that stability varies substantially across cases. For the aggregate any label, ConvNeXt-Base has the lowest mean Max-Sensitivity at 20.1052 ± 13.9192 , but that large SD means the average hides substantial case-level instability. Overall, the $n = 50$ sample is enough to show useful patterns, but it is not large enough to justify treating every mean difference as a settled ranking, especially without confidence intervals or paired statistical comparisons.

D. Bone-Window Shortcut Analysis

ResNet-50: The bone-channel analysis was rerun using the current Grad-CAM pipeline on the existing epoch-14 checkpoint, with no retraining. Sampling used 30 confirmed true positives per class, matching the ViT-B/16 and ConvNeXt-Base protocol, for 180 sampled observations in total (179 finite, 1 NaN in the epidural class).

TABLE VII Bone-Channel Correlation — ResNet-50 ($n = 30$ per Class)

Class	Total n	Finite n	Mean r	SD	Min	Max	$ r > 0.2$	$ r > 0.3$
any	30	30	0.0855	0.0748	-0.0466	0.2497	2/30	0/30
epidural	30	29	0.0365	0.0799	-0.1659	0.1753	0/30	0/30
intraparenchymal	30	30	0.0613	0.0482	-0.0127	0.1622	0/30	0/30
intraventricular	30	30	0.1010	0.0626	-0.0026	0.2306	2/30	0/30
subarachnoid	30	30	0.1149	0.0630	-0.0188	0.2867	2/30	0/30
subdural	30	30	0.1233	0.0768	-0.0490	0.2562	6/30	0/30

No ResNet-50 observation exceeds $|r| = 0.3$; the largest absolute correlation is 0.2867 (subarachnoid). Subdural has the most elevated observations, at 6/30 exceeding $|r| = 0.2$. The main limitation here is the sample size: 30 cases per class

gives a useful check for obvious alignment with the bone channel, but it cannot rule out rarer or more complicated shortcut behaviour.

ViT-B/16: $n = 30$ per class, 180 finite observations.

TABLE VIII Bone-Channel Correlation — ViT-B/16 ($n = 30$ per Class)

Class	Mean r	SD	Min	Max	$ r > 0.2$	$ r > 0.3$
any	-0.0490	0.2507	-0.4413	0.6609	13/30	8/30
epidural	-0.0141	0.2742	-0.4390	0.5341	17/30	10/30
intraparenchymal	-0.0111	0.1148	-0.3170	0.2993	3/30	1/30
intraventricular	0.1739	0.1743	-0.1843	0.4889	13/30	7/30
subarachnoid	0.0309	0.1934	-0.4453	0.3977	9/30	4/30
subdural	0.0558	0.2097	-0.3063	0.4557	12/30	5/30

ConvNeXt-Base: $n = 30$ per class, 180 finite observations.

TABLE IX Bone-Channel Correlation — ConvNeXt-Base ($n = 30$ per Class)

Class	Mean r	SD	Min	Max	$ r > 0.2$	$ r > 0.3$
any	0.0767	0.0811	-0.1095	0.2209	2/30	0/30
epidural	0.0819	0.1188	-0.2278	0.3550	5/30	1/30
intraparenchymal	0.0641	0.0466	-0.0166	0.2047	1/30	0/30
intraventricular	0.0823	0.0580	-0.0387	0.2015	1/30	0/30
subarachnoid	0.0857	0.0746	-0.0469	0.2362	1/30	0/30

subdural	0.1068	0.0686	-0.0878	0.2558	2/30	0/30
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All three backbones now use the same 30-per-class sampling design, so the results are directly comparable at the level of this check. ResNet-50 and ConvNeXt-Base both show modest mean correlations and relatively small SDs, with no observation exceeding $|r| = 0.3$. ViT-B/16 is different. Its class means are not uniformly high, and some are close to zero, but the SDs and threshold counts show much larger case-level variation. For example, the ViT-B/16 epidural mean is -0.0141 with an SD of 0.2742 , while 17/30 observations exceed $|r| = 0.2$ and 10/30 exceed $|r| = 0.3$. The mean alone would hide that variation.

Correlation is only a spatial association between the CAM and the bone channel. A high value does not prove the model used bone as a causal shortcut, and a low value does not prove the absence of more indirect bone dependence. The result is therefore best interpreted as a warning signal about the consistency of ViT-B/16 explanations, rather than as proof that it learned a bone shortcut.

E. Calibration

Temperature parameters fit on validation were 1.0439 for ResNet-50, 1.1364 for ViT-B/16, and 1.0932 for ConvNeXt-Base.

TABLE X Any-Level Calibration, Before $\rightarrow$ After Temperature Scaling

Backbone	ECE	MCE	Brier	NLL
ResNet-50	0.0296 $\rightarrow$ 0.0284	0.1160 $\rightarrow$ 0.1156	0.0863 $\rightarrow$ 0.0862	0.2838 $\rightarrow$ 0.2825
ViT-B/16	0.0204 $\rightarrow$ 0.0109	0.0743 $\rightarrow$ 0.0544	0.0772 $\rightarrow$ 0.0767	0.2587 $\rightarrow$ 0.2548
ConvNeXt-Base	0.0340 $\rightarrow$ 0.0297	0.1635 $\rightarrow$ 0.1308	0.0823 $\rightarrow$ 0.0818	0.2705 $\rightarrow$ 0.2669

All three models improve on every any-level metric after scaling, and ViT-B/16 shows the largest relative gain in ECE and MCE. Its post-scaling any-level ECE and MCE are also lower than the corresponding values for ResNet-50 and ConvNeXt-Base, which gives ViT-B/16 the clearest aggregate calibration result. The improvement is not uniform across labels, though, and that matters because a single temperature parameter is fit to recalibrate the model globally. A global improvement can coexist with a worse-calibrated subtype.

Epidural is the exception. ECE worsens after scaling for all three models: from 0.001572 to 0.003134 for ResNet-50, from 0.003907 to 0.007829 for ViT-B/16, and from 0.001701 to 0.004292 for ConvNeXt-Base. ViT-B/16's MCE also worsens, from 0.148 to 0.188, while the other two improve. Temperature scaling is therefore useful for the aggregate output, but it does not guarantee better probability estimates for every haemorrhage subtype. For a later PMCT evaluation, calibration should be reported at the subtype level rather than relying on the aggregate improvement as evidence that all model probabilities became more reliable.

F. Subtype and Cross-Backbone Synthesis

Epidural haemorrhage is the rarest class in this test set and shows the widest architecture-dependent spread across several metrics, which makes it a useful stress case for the analysis. It does not, by itself, establish that epidural should receive the greatest weight in a forensic PMCT decision. The eventual PMCT subtype distribution is unknown, and rarity in the RSNA test set is not a sufficient application-level weighting rule.

TABLE XI Epidural Haemorrhage Summary Across Backbones

Metric	ResNet-50	ViT-B/16	ConvNeXt-Base
ROC-AUC	0.8703	0.9361	0.9291
PR-AUC	0.0944	0.3970	0.2513
F1	0.1764	0.4144	0.3136
AOPC	0.114 $\pm$ 0.059	0.505 $\pm$ 0.206	0.338 $\pm$ 0.138
Max-Sens.	32.60	29.94	18.22

The broader pattern is more useful than a single epidural ranking. ConvNeXt-Base leads on intraparenchymal and intraventricular AOPC and has the lowest Max-Sensitivity across all six classes (any and every haemorrhage subtype), with particularly strong intraparenchymal stability. ViT-B/16 leads on epidural AOPC, macro ROC-AUC, macro PR-AUC, and aggregate calibration improvement, although its bone-correlation results also show the largest case-level variation. ResNet-50 remains competitive on intraventricular AOPC, but its epidural classification and AOPC results are clearly the weakest of the three. Which architecture looks strongest therefore depends on which subtype and evaluation dimension is considered.

An anatomical interpretation provides a plausible mechanism for these differences, although it remains a hypothesis rather than a demonstrated causal explanation. Epidural haemorrhage is an extra-axial collection between the dura and inner table of the skull, and it commonly has a lentiform or biconvex appearance because its lateral spread is constrained by cranial sutures [14]. The discriminative signal is therefore concentrated at the skull-brain interface. ViT-B/16 divides the image into 16×16 patches and applies self-attention across the patch tokens from the first transformer block onward [7]. Global token interactions could make peripheral skull-boundary features influential

even when the haemorrhage itself is small. This is consistent with the diffuse or peripheral CAMs and the larger case-level variation in the bone-channel correlations observed for ViT-B/16. The present experiments did not manipulate the attention mechanism or skull features, however, so they cannot establish that self-attention caused the observed pattern.

For intraparenchymal and intraventricular haemorrhage, the target pathology is located within localised soft-tissue or ventricular structures. ConvNeXt-Base uses 7×7 depthwise convolutions to enlarge local receptive fields while retaining the spatial inductive bias of convolutional processing [8]. That combination provides a plausible explanation for its stronger AOPC and Max-Sensitivity results on intraparenchymal and intraventricular haemorrhage. The claim is specifically about convolutional locality and receptive-field structure; exact translation equivariance of the entire ConvNeXt network is not assumed, since normalisation and downsampling can modify that property.

VI. DISCUSSION

The main finding is that architecture changes explanation behaviour in ways that do not simply follow classification performance. ConvNeXt-Base has the strongest AOPC and Max-Sensitivity results for intraparenchymal haemorrhage and is also strong on intraventricular haemorrhage. ViT-B/16 has the best aggregate classification metrics and the strongest epidural AOPC. ResNet-50 is clearly weakest on epidural classification and epidural AOPC. Classification performance therefore does not predict explanation behaviour uniformly across subtypes.

There is also a methodological reason to be careful about calling these architecture effects. The comparison is not a controlled ablation. Optimizer, learning rate, and warmup schedule differ between ResNet-50 and the other two backbones, and fine-tuned depth differs across all three. Some of ResNet-50's weaker epidural behaviour could therefore come from its training configuration rather than its architecture. The same limitation applies to any explanation-quality difference attributed directly to architecture. The data show that the complete configurations behave differently; they do not isolate why.

ViT-B/16 is the clearest example of why the explanation metrics need to be read alongside the maps themselves. Its epidural AOPC lead over the other two backbones (Section V-C, Table XI) is real, yet several displayed CAMs are diffuse or peripheral, and the bone-correlation analysis shows much greater case-level variation than the other two backbones. The discrepancy highlights a divergence between perturbation sensitivity and qualitative localisation.

No single architecture uniformly outperforms the others across all metrics; performance varies by haemorrhage

subtype and evaluation dimension. ViT-B/16 is strongest on aggregate classification and epidural AOPC, whereas ConvNeXt-Base is stronger for intraparenchymal and intraventricular explanation behaviour. Treating epidural as the deciding class simply because it is rare in RSNA would introduce an application-level weighting that was not defined in this study.

For the next stage, ViT-B/16 and ConvNeXt-Base should be carried forward as complementary candidates. ViT-B/16 has the stronger aggregate classification results and the strongest epidural AOPC. ConvNeXt-Base has stronger explanation faithfulness and stability for intraparenchymal and intraventricular haemorrhage, along with less case-level variation in the bone-correlation check. ResNet-50 has a clear weakness on epidural classification and AOPC, and it does not have a compensating advantage large enough in the current results to make it a primary candidate. If the PMCT study eventually has to select one model, that choice should be made after defining which clinical or forensic outcomes are being prioritised, rather than treating this clinical baseline as if it already supplies the weighting.

VII. FORENSIC APPLICATION AND IMPLICATIONS

The intended role here is decision support, not automated diagnosis. A classification paired with a spatially plausible explanation could help a forensic reviewer direct attention towards a candidate region for closer inspection against the underlying CT. If these properties hold up on PMCT, the practical benefit would be a more inspectable model and potentially less review burden. Because antemortem clinical CT serves as the training domain, direct performance on post-mortem CT (PMCT) remains to be evaluated. Nothing in the present study establishes legal evidentiary validity.

VIII. LIMITATIONS

- Not a controlled ablation. Optimizer, learning rate, warmup, and fine-tuned depth differ across configurations, so differences cannot be attributed to architecture alone.
- No statistical inference. No confidence intervals or significance tests are reported for any metric; all comparisons are point estimates. The AOPC and Max-Sensitivity samples are $n = 50$ per class, and the bone-correlation checks are $n = 30$ per class. Those sample sizes are enough to expose broad differences, but the SDs show substantial case-level variation, so close mean differences should not be treated as settled rankings.
- Independence assumptions are unaddressed. Whether multiple slices per patient or case create correlated observations is not resolved in the source material, and the effective amount of independent evidence could be smaller than the slice counts suggest.

- Qualitative Grad-CAM examples are illustrative, not population-level evidence. They can show concrete failure modes and tensions with the quantitative metrics, but they cannot establish how often those patterns occur.
- No ground-truth localisation annotations were available, so anatomical correctness and causal shortcut use were not directly evaluated.
- No PMCT data are included. Everything above is clinical/source-domain only, so the central question of whether these explanation properties survive domain shift remains unanswered.

IX. FUTURE WORK: PMCT DOMAIN-SHIFT EVALUATION

The necessary next stage is a clinical-to-PMCT transfer evaluation, which has not yet been performed. Matsumoto et al. [3] establish that clinical-to-PMCT classification transfer is feasible; what remains open is whether the explanation behaviour observed on clinical CT survives the same transfer. Both ViT-B/16 and ConvNeXt-Base should be evaluated, because their strengths differ across classification and explanation metrics.

PMCT introduces physical changes that are absent or uncommon in antemortem clinical CT, including internal hypostasis and dependent blood sedimentation, loss of grey-white matter differentiation, putrefactive gas, cerebral settling or collapse, and later liquefaction of the brain parenchyma [15, 16]. These changes can alter local attenuation patterns and the broader spatial context used by the classifier and Grad-CAM. The working hypothesis is that ConvNeXt-Base will be more resilient to diffuse post-mortem artefact changes, because its convolutional inductive bias emphasises local spatial structure. ViT-B/16, in contrast, may be more sensitive to broad redistribution of intensity or peripheral context, since self-attention integrates information across the full patch sequence. ViT-B/16 may still benefit from global context when artefact and haemorrhage interact over larger spatial scales, so both models should be evaluated rather than assuming that one inductive bias will dominate.

Dataset: PMCT case/slice counts, positive/negative counts, subtype counts, source and acquisition characteristics, any overlap with RSNA identifiers, and whether labels are case-, slice-, or region-level and expert-confirmed.

Transfer protocol: Whether backbones are applied to PMCT with or without fine-tuning; if fine-tuned, on what data; whether PMCT labels inform threshold selection; whether temperature scaling is refit; whether the evaluation set is fully held out.

Explainability: Grad-CAM behaviour, AOPC where applicable, Max-Sensitivity, bone-channel correlation, and qualitative examples and failures, for each backbone.

Central comparison: Does the architecture-dependent explanation behaviour seen here on clinical data persist under domain shift? This matters to the eventual application, because classification transfer alone would not establish whether the explanations remain useful.

Statistical rigour should also improve at this stage: confidence intervals for classification metrics, bootstrap intervals for AOPC and Max-Sensitivity, formal comparison between backbones rather than point-estimate ranking, case-level rather than slice-level independence handling, and treatment of any small annotated PMCT subset as a validation set rather than an assumption of full pixel-level supervision.

X. CONCLUSION

This report establishes a clinical-CT baseline for ResNet-50, ViT-B/16, and ConvNeXt-Base on ICH detection, with particular attention to the faithfulness and stability of their explanations. The main result is not that one backbone wins. It is that architecture changes the balance between classification and explanation behaviour. ViT-B/16 has the strongest aggregate classification results and the strongest epidural AOPC, while ConvNeXt-Base has the strongest intraparenchymal and intraventricular explanation results and generally lower Max-Sensitivity. ResNet-50 is clearly weakest on epidural classification and AOPC. These are real differences, but the SDs, the small n used for explanation evaluation, and the lack of statistical inference mean that close point estimates should not be turned into a formal ranking.

ViT-B/16 and ConvNeXt-Base should therefore be carried into the PMCT domain-shift evaluation. A pre-specified weighting, if one model must eventually be chosen, should be defined before that decision is made, so that the PMCT experiment can determine whether one has a practical advantage. None of this establishes forensic utility on its own. Whether the classification and explanation properties seen on clinical CT survive the clinical-to-PMCT domain shift remains the central question for the next stage.

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